

AUTOMATED NETWORK REQUEST MANAGEMENT

ServiceNow Custom Application

Complete Project Documentation

Platform: ServiceNow

Project: Automated Network Request Management

Scope: Service Catalog, Approvals and Flow Designer Automation

Document: Implementation, Catalog, Automation and Testing Report

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1. Project Overview

The Automated Network Request Management project is a ServiceNow solution that lets employees and network teams submit, route, approve, and fulfil network-related service requests through a structured catalog record and Service Catalog experience. The implementation combines a custom network request table, catalog-driven form design, conditional UI Policies, an approval relationship, and Flow Designer automation.

- Requesters submit network requests (connections, access, devices) through a guided catalog form.
- Requests are stored in a custom network request table alongside standard platform tables.
- Conditional UI Policies reveal additional fields based on selections (e.g. connection type, device type).
- An Approval Request relationship links approval records (sysapprovals) to each network request.
- Flow Designer automates record creation, approval routing, email notification, and status updates.
- List views give approvers and network teams centralized visibility into requests, tasks, and statuses.

2. Problem Statement and Objectives

Network teams need a simple, consistent method to receive and process network-related service requests. A structured ServiceNow solution reduces unorganized, manual requests and gives network teams a consistent record containing requester and request information.

Objectives

- Streamline and automate the end-to-end lifecycle of network-related service requests.
- Reduce manual effort and human error in request intake and routing.
- Accelerate request fulfilment times.
- Enforce standardized workflows and approval processes.
- Improve visibility and tracking through a centralized platform.
- Enhance end-user experience with faster, more reliable service delivery.
- Ensure compliance with IT and security policies.

3. Application and Catalog Item Creation

The functional scope covers catalog creation, form design, approval routing, Flow Designer automation, and email notification. A Network Request catalog item is configured within the Service Catalog, mapped to the custom network request table, with stakeholder mapping and an execution roadmap defined during the design phase.

The screenshot displays the ServiceNow 'Catalog Item - Network Request' configuration page. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and a search bar. The main header shows 'Catalog Item - Network Request' with a star icon and a search bar. Below the header, a blue informational box states: 'Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies. • Enter a Name and Short description to display for the item. • Enter a Price, approvals, variables, and other information as needed.' The form fields are organized into two columns. The left column contains: 'Name' (text field with 'Network Request'), 'Catalogs' (dropdown with 'Service Catalog'), 'Category' (text field with 'Network Standard Changes'), 'State' (dropdown with '-- None --'), 'Checked out' (dropdown with '-- None --'), and 'Owner' (text field with 'System Administrator'). The right column contains: 'Application' (dropdown with 'Global'), 'Active' (checkbox checked), and 'Fulfillment automation level' (dropdown with 'Unspecified'). Below the form fields, a tabbed interface shows 'Item Details' (selected), 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'. The 'Item Details' tab has a 'Short description' field with 'Network Services Request' and a 'Description' field with a rich text editor. The rich text editor toolbar includes bold, italic, underline, link, unlink, text color, background color, bulleted list, numbered list, indent, outdent, and a help icon.

The screenshot shows the 'Catalog Item - Network Request' form in ServiceNow. At the top, there's a navigation bar with 'All', 'Favorites', 'History', and 'Workspaces'. The main header is 'Catalog Item - Network Request'. Below this, there's a search bar and a 'Meta' field with a placeholder text: 'Add relevant tags to the Meta field using comma-separated list of tags. These tags will be used while searching the item. Not applicable if AI Search is configured.' Below the 'Meta' field, there are buttons for 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. A section titled 'related Links' contains links for 'em Diagnostic' and 'un Point Scan'. Below this, there's a tabbed interface with tabs for 'Variables (7)', 'Variable Sets (1)', 'Catalog UI Policies (1)', 'Catalog Client Scripts', 'Available For', 'Not Available For', 'Categories (1)', 'Catalogs (1)', 'Catalog Data Lookup Definitions', 'Related Articles', and 'Related Catalog Items'. The 'Assigned Topics' section is also visible. At the bottom, there's a table with columns: 'Short description', 'Variable set', 'Conditions', 'Reverse if false', 'On load', 'Inherit', 'Updated', and 'Order'. The table contains one row with the following data: 'visible/hide of ID field', '(empty)', true, true, false, 2026-09-04 04:53:24, 100.

Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
visible/hide of ID field	(empty)		true	true	false	2026-09-04 04:53:24	100

4. Custom Table and Data Architecture

The Automated Network Request Management solution is built on a structured data model that combines standard platform tables with a custom network request table for clarity, scalability, and control. Network requests are captured through catalog-driven records and stored with well-defined fields such as request type, access level, business justification, and related network configuration items.

Forms are designed to dynamically display only relevant information using UI logic, while mandatory fields enforce data accuracy and compliance. List views provide operational visibility for approvers and network teams, enabling efficient tracking of requests, tasks, and statuses. This architecture ensures secure access, seamless automation, and reliable reporting across the entire request lifecycle.

The screenshot shows the 'Table - Database Table' form in ServiceNow. At the top, there's a navigation bar with 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The main header is 'Table - Database Table'. Below this, there's a search bar and a 'Delete' button. A section titled 'A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)' contains a 'Label' field with the value 'Database Table', an 'Application' field with the value 'Global', and a 'Name' field with the value 'u_database_table'. Below this, there's a 'Remote Table' field. A tabbed interface is visible with tabs for 'Columns', 'Controls', 'Application Access', and 'Prescription'. The 'Columns' tab is selected, showing a table with columns: 'Column label', 'Type', 'Reference', 'Max length', 'Default value', and 'Display'. The table contains the following data:

Column label	Type	Reference	Max length	Default value	Display
Total Amount	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Address	String	(empty)	40		false
Created by	String	(empty)	40		false
Mode of Payment	String	(empty)	40		false
Type of Connection	String	(empty)	40		false

The screenshot displays the ServiceNow interface for configuring a table. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The main header shows 'Table - Database Table' with a search bar and action buttons: 'Delete', 'Update', and 'Delete All Records'.

The table configuration section lists the following fields:

Field Name	Type	Value	Length	Script	Boolean
Type of Connection	String	(empty)	40		false
AssignedTo	Reference	User	32		false
Approval Status	String	(empty)	40		false
Database Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();	false
Created	Date/Time	(empty)	40		false
Updated by	String	(empty)	40		false
Requested For	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Mobile Number	String	(empty)	40		false

Below the table configuration, there are buttons for 'Delete', 'Update', and 'Delete All Records'. A 'Related Links' section provides links to 'Form Builder', 'Design Form', 'Layout Form', 'Layout List', 'Show Form', 'Show List', 'Show Schema Map', 'Add to Service Catalog', 'Run Point Scan', and 'Explore REST API'.

The second screenshot shows the 'Database Table - DAT00001010' form. It includes fields for 'Database Number' (DAT00001010), 'Mobile Number' (909090909), 'Total Amount' (800), 'Mode of Payment' (card), 'Requested For' (Sampath), 'Type of Connection' (existing), 'Address' (Hyderabad), 'Assignment To' (ITIL User), and 'Approval Status' (Approved). Buttons for 'Update' and 'Delete' are visible.

Below the form, there is a section for 'Approvals' with a table showing the approval status:

State	Approver	Comments	Approval for	Created
Approved	(empty)		(empty)	2026-09-04 03:06:46

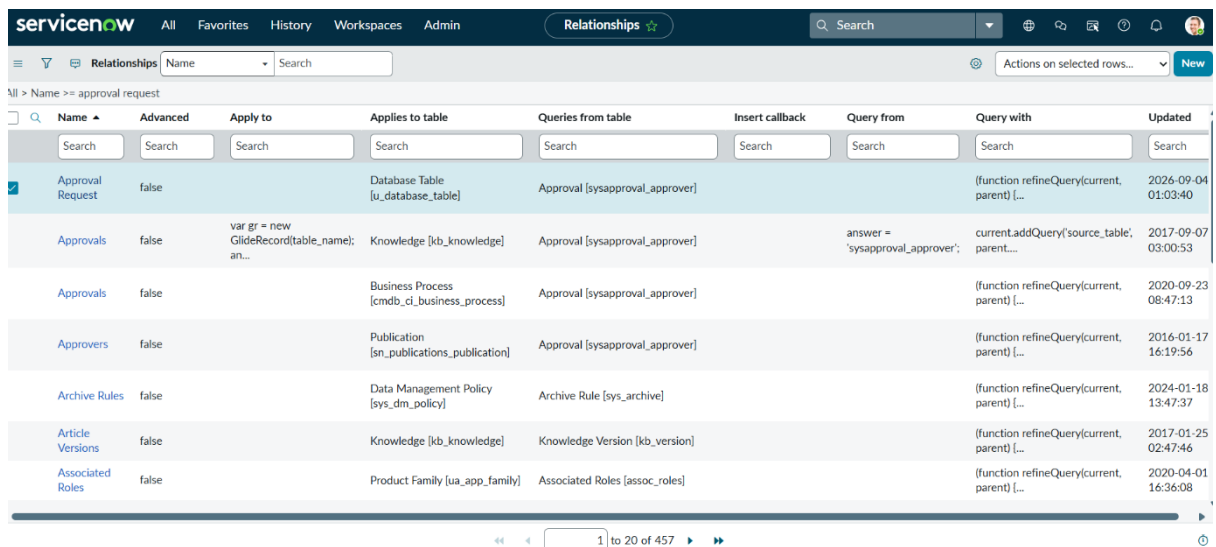
The bottom of the page shows a pagination bar indicating '1 to 1 of 1'.

5. Approval Request Relationship

Creation of Approval Request on the Network Database Table

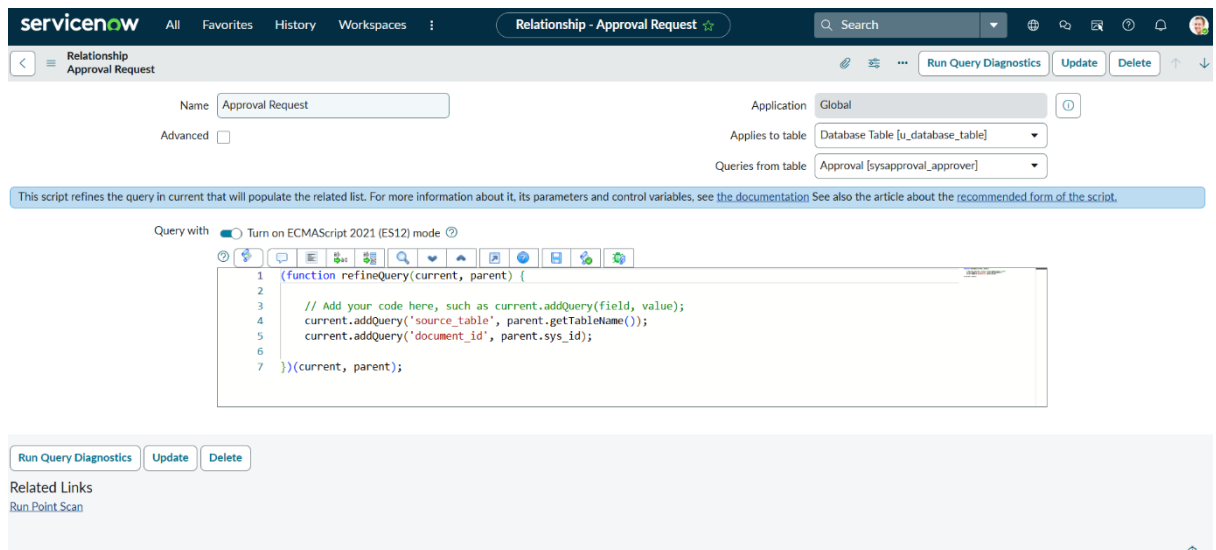
Activity 1: Creation of Related List

- Navigate to System Definition > Relationships.
- Click New to create a new relationship.
- Name: Approval Request.
- Applies to Table: the custom network request (database) table.
- Queries from Table: the sysapprovals table.
- Active: set to True.
- Save the relationship.



The screenshot shows the ServiceNow 'Relationships' list view for 'Approval Request'. The table lists various relationships with columns for Name, Advanced, Apply to, Applies to table, Queries from table, Insert callback, Query from, Query with, and Updated. The first row is selected.

Name	Advanced	Apply to	Applies to table	Queries from table	Insert callback	Query from	Query with	Updated
Approval Request	false		Database Table [u_database_table]	Approval [sysapproval_approver]			(function refineQuery(current, parent) [...	2026-09-04 01:03:40
Approvals	false	var gr = new GlideRecord(table_name); an...	Knowledge [kb_knowledge]	Approval [sysapproval_approver]		answer = 'sysapproval_approver';	current.addQuery('source_table', parent...	2017-09-07 03:00:53
Approvals	false		Business Process [cmdb_ci_business_process]	Approval [sysapproval_approver]			(function refineQuery(current, parent) [...	2020-09-23 08:47:13
Approvers	false		Publication [sn_publications_publication]	Approval [sysapproval_approver]			(function refineQuery(current, parent) [...	2016-01-17 16:19:56
Archive Rules	false		Data Management Policy [sys_dm_policy]	Archive Rule [sys_archive]			(function refineQuery(current, parent) [...	2024-01-18 13:47:37
Article Versions	false		Knowledge [kb_knowledge]	Knowledge Version [kb_version]			(function refineQuery(current, parent) [...	2017-01-25 02:47:46
Associated Roles	false		Product Family [ua_app_family]	Associated Roles [assoc_roles]			(function refineQuery(current, parent) [...	2020-04-01 16:36:08



The screenshot shows the ServiceNow 'Relationship - Approval Request' configuration page. It includes fields for Name, Application, Applies to table, and Queries from table. A code editor shows a JavaScript function for refining the query. Below the code editor are buttons for 'Run Query Diagnostics', 'Update', and 'Delete'. A 'Related Links' section is also present.

Name: Approval Request

Application: Global

Applies to table: Database Table [u_database_table]

Queries from table: Approval [sysapproval_approver]

This script refines the query in current that will populate the related list. For more information about it, its parameters and control variables, see [the documentation](#). See also the article about the [recommended form of the script](#).

Query with: ☐ Turn on ECMAScript 2021 (ES12) mode

```
1 (function refineQuery(current, parent) {
2
3   // Add your code here, such as current.addQuery(field, value);
4   current.addQuery('source_table', parent.getTabletime());
5   current.addQuery('document_id', parent.sys_id);
6
7 })(current, parent);
```

Run Query Diagnostics Update Delete

Related Links

[Run Point Scan](#)

This relationship surfaces approval records directly on the network request form as a related list, giving requesters and approvers a single place to track approval status.

6. Form Design and Field Configuration

The Network Request form is designed to collect the information required to process a request, separating system-generated information (such as request number) from user-entered or selected information (such as request type, access level, business justification, and related network configuration items).

Requested Item - RITM0010003

Number: RITM0010003

Item: Network Request

Request: REQ0010003

Requested for: System Administrator

Due date: 2026-09-12 05:59:43

Configuration item:

Watch list:

Opened: 2026-09-10 05:59:43

Opened by: System Administrator

Stage: Assess or Scope Task

State: Open

Quantity: 1

Estimated delivery:

Backordered: ☐

Order Guide:

Requester Information

Opened on behalf of: Abraham Lincoln

User Name: Abraham Lincoln

Email ID:

Phone Number:

Activities: 1

System Administrator

Impact: 3 - Low

Opened by: System Administrator

Priority: 4 - Low

State: Open

Field changes: 2026-09-10 05:59:43

Update Delete

Catalog Tasks (2) Approvers Group approvals

Number Search

Request item = RITM0010003

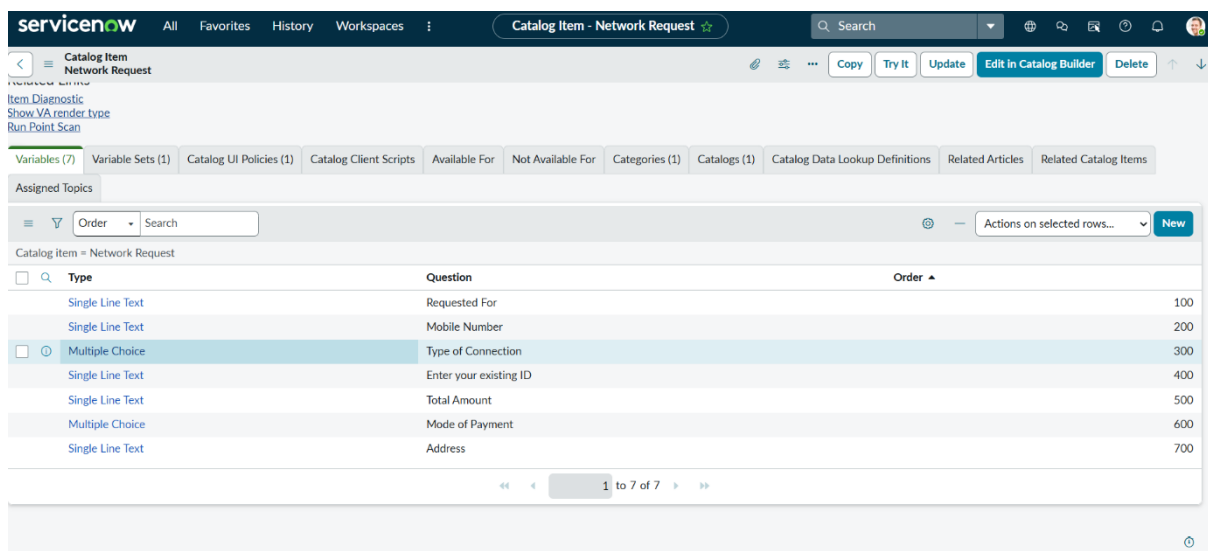
Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order
SCTASK0010006	Software	(empty)	Provide requested service	(empty)	(empty)	200
SCTASK0010005	Field Services	(empty)	Assess or Scope Task	2026-09-10 05:59:45	(empty)	100

1 to 2 of 2

7. Catalog Variables Configuration

Variables are configured on the Network Request catalog item to build the form fields presented to the requester.

- Variable types include Single Line Text, Multi Line Text, Reference, and Choice, selected per requirement.
- Order values (100, 200, 300, ...) control the display sequence of fields.
- Question: the label shown to the requester.
- Name: the internal variable name used for scripting.
- Tooltip: guidance shown when the cursor hovers over the field.
- Example text: sample input to guide the requester.
- Mandatory / Read-only: configured as required per field.
- Auto-populate: dependent variables use dot-walking to pull related values automatically.

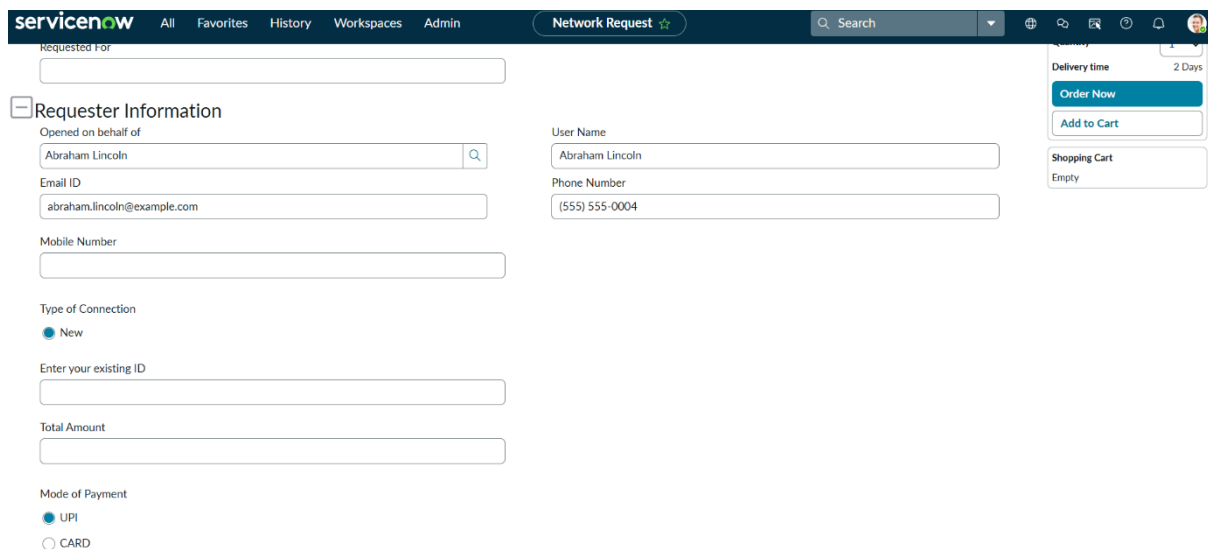


Type	Question	Order
Single Line Text	Requested For	100
Single Line Text	Mobile Number	200
Multiple Choice	Type of Connection	300
Single Line Text	Enter your existing ID	400
Single Line Text	Total Amount	500
Multiple Choice	Mode of Payment	600
Single Line Text	Address	700

8. Navigation Flow

Procedure:

- Log in to the ServiceNow instance.
- Copy the instance domain, e.g. <https://dev190678.service-now.com>.
- Append the /sp prefix to reach the Service Portal, e.g. <https://dev190678.service-now.com/sp>.
- Search for "Network Requests".
- Fill in the required details and click Submit.
- A new request is generated with a request number, and the requester receives a confirmation email.



Requested For

Requester Information

Opened on behalf of

Abraham Lincoln

Email ID

abraham.lincoln@example.com

Mobile Number

Type of Connection

New

Enter your existing ID

Total Amount

Mode of Payment

UPI

CARD

User Name

Abraham Lincoln

Phone Number

(555) 555-0004

Delivery time

2 Days

Order Now

Add to Cart

Shopping Cart

Empty

9. UI Policy and Conditional Validation

Catalog UI Policy Configuration

UI Policies display additional fields when certain choices are made, using conditional logic and field dependencies configured within ServiceNow.

Scenario 1: If the user selects "types of Connection" = Existing, the Existing ID field is displayed. Scenario 2: If the user selects "types of devices" = Others, the "Please specify" field is displayed.

Procedure:

- Navigate to the Network Request catalog item.
- Open the Catalog UI Policy related list and click New.
- Set Applies to = Catalog item, and Catalog item = Network Request.
- Provide a short description and the triggering condition (e.g. types of Connection is Existing).
- Save, then configure the UI Policy Action for the target variable and set it Visible = True.
- Update the UI Policy and test it on the catalog form.

The screenshot shows the ServiceNow interface for configuring a Catalog UI Policy. The title bar indicates 'Catalog UI Policy - visible/hide of ID field'. The main configuration area includes several checkboxes: 'Applies on a Catalog Item view' (checked), 'Applies on Catalog Tasks' (checked), and 'Applies on Requested Items' (unchecked). There are two blue informational boxes: one stating 'Apply the catalog UI policy actions when the form is loaded or when the user changes values on the form' and another stating 'Reverse the effects of the catalog UI policy actions when the Conditions evaluate to false'. Below these, there are 'Update' and 'Delete' buttons. A 'Related Links' section contains a link to 'Run Point Scan'. At the bottom, there is a table titled 'Catalog UI Policy Actions' with columns for Name, Read only, Mandatory, Visible, and Order. The table contains one row with the name 'enter_your_existing_id', 'Read only' set to 'Leave alone', 'Mandatory' set to 'Leave alone', 'Visible' set to 'True', and 'Order' set to '100'. A pagination bar at the bottom indicates '1 to 1 of 1'.

10. Automation — Flow Designer

Flow Designer logic implements: Get Catalog Variables, Create Record, Ask for Approval, Send Email, Update Record — triggered on catalog item submission and supporting dynamic automation.

Activity 1: Creation of Flow

- Navigate to the Flow Designer home page.
- Click New to create a new flow.
- Provide the flow name: Network Request.
- Provide a description of the flow.
- Click Build flow, then add the Get Catalog Variables, Create Record, Ask for Approval, Send Email, and Update Record actions in sequence.

The screenshot shows the 'EXECUTION DETAILS' for a flow named 'Network Request'. The flow is currently in a 'Waiting' state. The table below lists the actions and their execution status.

Step	Action	Type	State	Start time	Duration
1	Get Catalog Variables from Network Request	Core Action	Completed	2026-09-10 06:29:55	20ms
2	Create Record	Core Action	Completed	2026-09-10 06:29:55	6ms
3	Send Email	Core Action	Completed	2026-09-10 06:29:55	28ms
4	Ask For Approval	Core Action	Waiting	2026-09-10 06:29:55	55ms
5	If If Request is Approved	Flow Logic	Not Run		
6	Update Record	Core Action	Not Run		
7	Else	Flow Logic	Not Run		
8	Update Record	Core Action	Not Run		

Step	Action	State	Start time	Duration
1	Get Catalog Variables from Network Request	Completed	2026-09-10 06:29:55	20ms
2	Create Record	Completed	2026-09-10 06:29:55	6ms
3	Send Email	Completed	2026-09-10 06:29:55	28ms
4	Ask For Approval	Completed	2026-09-10 06:29:55	144ms
5	If Request is Approved	Evaluated - True	2026-09-10 06:34:23	8ms
6	Update Record	Completed	2026-09-10 06:34:23	8ms
7	Else	Not Run		
8	Update Record	Not Run		

11. Access Control

Role-based access control restricts who can view, create, and update network request records and approvals, ensuring that only authorized requesters, approvers, and network team members can act on a given request.

State	Approver	Comments	Approval for	Created
Approved	System Administrator		(empty)	2026-09-13 02:30:54
Approved	System Administrator		(empty)	2026-09-10 06:29:55
Approved	System Administrator		(empty)	2026-09-13 00:18:37
Approved	System Administrator		(empty)	2026-09-13 02:04:42
Approved	System Administrator		(empty)	2026-09-13 02:29:30
Approved	System Administrator		(empty)	2026-09-13 03:38:30
Approved	System Administrator		(empty)	2026-09-06 06:19:34
Approved	System Administrator		(empty)	2026-09-10 06:29:48

Approval - Database Table: RITM0010008

Approver: System Administrator

State: Approved

Approval Reason:

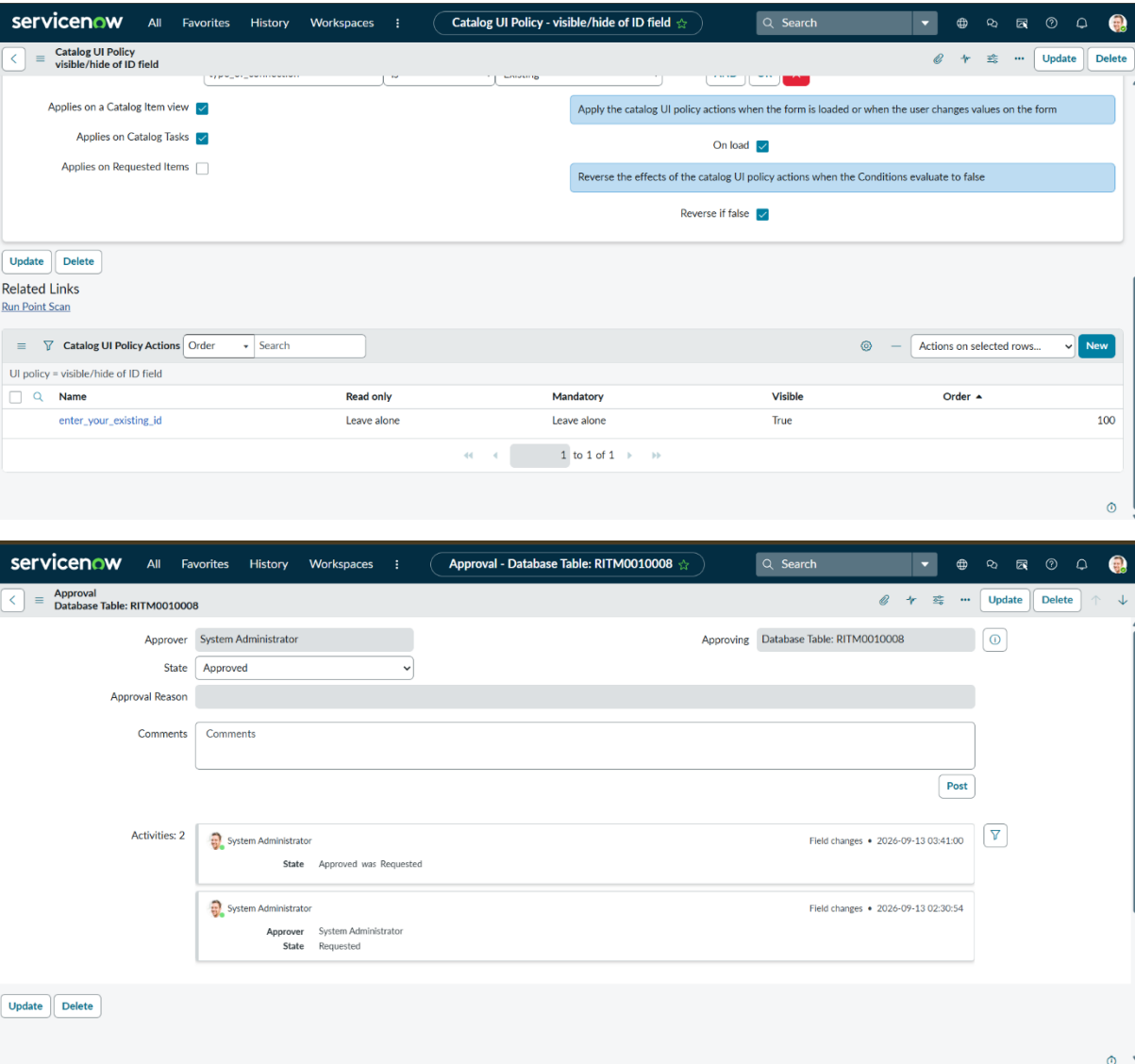
Comments:

Activities: 2

- System Administrator: State Approved was Requested (Field changes: 2026-09-13 03:41:00)
- System Administrator: Approver System Administrator, State Requested (Field changes: 2026-09-13 02:30:54)

12. QA Testing

Functional QA testing covered catalog submission, conditional UI Policy behavior, approval routing, automation execution, and record visibility for approvers and network teams.



13. Data Integrity

Mandatory fields, reference fields, and dependent/auto-populated variables enforce data accuracy across the request lifecycle, ensuring that records entering the network request table are complete and consistent.

The screenshot shows the ServiceNow interface for a Request record (REQ0010003). The form is divided into several sections:

- Header:** ServiceNow logo, navigation tabs (All, Favorites, History, Workspaces, Admin), and a search bar.
- Form Fields:**
 - Number: REQ0010003
 - Requested for: System Administrator
 - Location: (empty)
 - Due date: 2026-09-12 05:59:43
 - Price: \$0.00
 - Description: (empty)
 - Short description: (empty)
 - Special instructions: (empty)
 - Comments (Customer visible): (empty)
- Metadata:**
 - Opened: 2026-09-10 05:59:44
 - Opened by: System Administrator
 - Approval: Approved
 - Request state: Approved
- Buttons:** Discuss, Follow, Update, Cancel Request, Copy, Delete.
- Footer:** Activities: 1, System Administrator, Field changes: 2026-09-10 05:59:44.

14. Functional Flow



Figure 1: End-to-end functional flow — Requester to notification.

Step	Component	Function
1	Service Catalog	Requester submits the Network Request catalog item.
2	Catalog Variables	Capture request type, access level, justification and related items.
3	UI Policy	Display conditional fields based on connection/device type.
4	Custom Table	Store the network request record.
5	Approval Request Relationship	Link the request to its sysapprovals approval record.
6	Flow Designer	Create record, request approval, send email, update record status.
7	List Views	Give approvers and network teams visibility into requests and status.
8	Notification	Requester receives email confirmation and status updates.

End-to-end architecture: Requester → Service Portal → Network Request catalog item → Custom Table → Approval Request relationship → Flow Designer automation → Status-based tracking.

15. Testing and Validation

ID	Test	Expected / Evidence
TC-01	Submit network request	Request record is created in the custom network request table.
TC-02	Mandatory field validation	Form blocks submission until required fields are completed.
TC-03	Connection type = Existing	Existing ID field becomes visible per UI Policy.
TC-04	Device type = Others	"Please specify" field becomes visible per UI Policy.
TC-05	Approval relationship	Approval Request related list shows the linked sysapprovals record.
TC-06	Flow Designer execution	Get Catalog Variables, Create Record, Ask for Approval, Send Email and Update Record execute successfully on submission.
TC-07	Notification	Requester receives an email confirming submission.
TC-08	List view visibility	Approvers and network teams see the request and its current status.

Emails

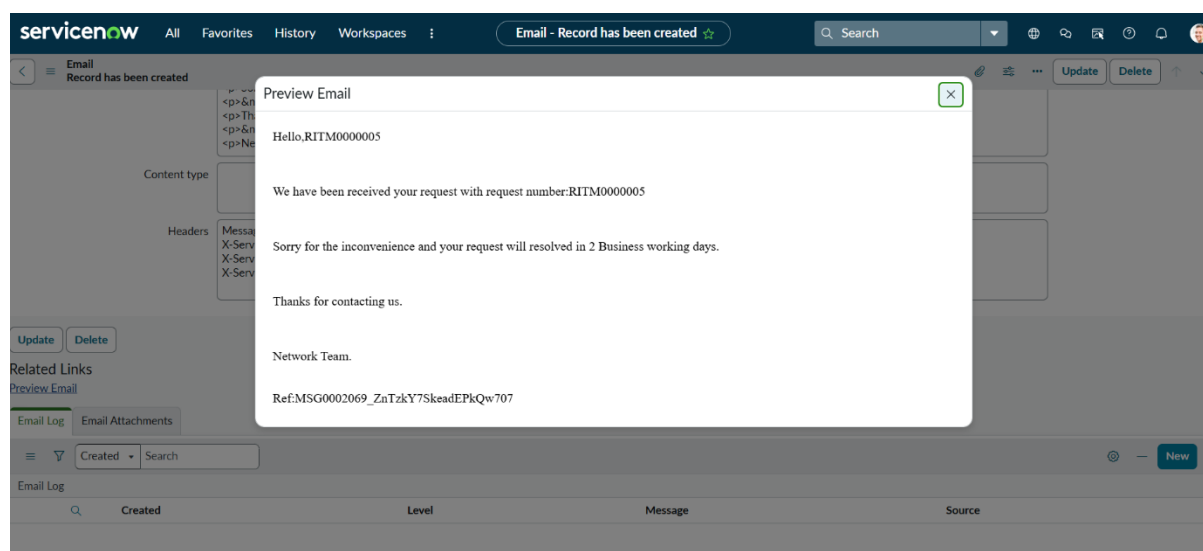
Created

Search

Actions on selected rows...

All > Created on Today

☐	Created	Recipients	Subject	Type	Notification type	User ID
☐	2026-09-02 20:59:51	abraham.lincoln@example.com	Employee Onboarding Request - HR Approval Required	send-ready	SMTP	(empty)
☒	2026-09-02 20:59:51	abraham.lincoln@example.com	Employee Onboarding Request Awaiting Your Approval	send-ready	SMTP	(empty)
☐	2026-09-02 01:00:10	aileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
☐	2026-09-02 01:00:09	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
☐	2026-09-02 20:58:40	abel.tuter@example.com	Employee Onboarding Request Awaiting Your Approval	send-ready	SMTP	(empty)
☐	2026-09-02 10:01:09	admin@example.com	[ITSM Analytics] Daily Data Collection completed with warning	send-ready	SMTP	(empty)
☐	2026-09-02 01:00:10	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
☐	2026-09-02 00:00:09	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)



A recorded walkthrough of the implementation is available at:

https://drive.google.com/file/d/1gDrM3CmFP848MBk_zRy7qVAJv1ECrgzL/view?usp=sharing

16. Conclusion and Future Enhancements

The Automated Network Request Management project demonstrates an end-to-end ServiceNow solution covering catalog-based intake, structured data storage, conditional form logic, approval tracking, and Flow Designer automation.

Key Achievements

- Structured data architecture combining standard and custom network request tables.
- Network Request catalog item with ordered, well-configured variables.
- Conditional UI Policies for connection type and device type scenarios.
- Approval Request relationship linking requests to sysapprovals records.
- Flow Designer automation covering the full request lifecycle.
- Role-based access control and validated data integrity.

Future Enhancements

- SLA-based escalation and automated reassignment for overdue requests.
- Deeper CMDB integration for network configuration items.
- Reporting dashboards for request volume, approval turnaround and fulfilment time.
- Expanded, multi-level approval hierarchies.
- Additional request types and knowledge-article recommendations during submission.

End of Project Documentation

Automated Network Request Management — ServiceNow